
Polynomial-Linked Tucker Rank Lifts for Nonlinear Tensor Observations

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Abstract

1 Classical Tucker models explain multiway data with a multilinear low-rank struc-
2 ture. This is effective when the observed tensor is close to a multilinear signal, but
3 many sensing and imaging pipelines include nonlinear entry-level effects, such as
4 response curves, saturation, and material-dependent distortions. These effects can
5 make a low-rank latent tensor appear higher-rank in the observed domain.

6 We propose Nonlinear Tucker Decomposition with Polynomial Links (NTD-PL),
7 a parameter-tied Tucker enhancement and shared-backbone rank lift for such
8 observations. NTD-PL first forms a shared Tucker latent tensor and then maps each
9 latent entry through a finite polynomial link. The model contains standard Tucker
10 as the degree-one case and adds only a small number of scalar link coefficients.
11 At the same time, powers of the latent tensor induce tied high-order interactions
12 among the same Tucker factors, giving a compact rank-lift without introducing a
13 full high-order tensor parameter.

14 We study the representation power of this model through approximation and sep-
15 aration results, and derive a masked learning objective for reconstruction and
16 completion. The resulting solver alternates between Tucker backbone updates and
17 a small ridge-regression step for the link coefficients, with degree continuation for
18 stability. Experiments on controlled tensors, CAVE hyperspectral completion and
19 reconstruction, and external HSI scenes show that NTD-PL improves matched-rank
20 Tucker when the residual has a coherent nonlinear component. Rank-lift and link-
21 yield diagnostics show that the gain is tied to jointly reshaping the Tucker backbone
22 under a shared nonlinear link, rather than to a small ordinary rank increase alone.

1 Introduction

24 Tensor decompositions provide compact models for multiway data in scientific sensing, imaging,
25 and signal processing [Kolda and Bader, 2009, Sidiropoulos et al., 2017, Liu et al., 2013]. Tucker
26 decomposition is a standard choice because it separates a small core from mode-wise factors, but
27 its observation model is linear in the reconstructed latent entry. This can be too restrictive when a
28 low-rank latent signal passes through saturation, response curves, calibration effects, or material-
29 dependent nonlinearities before it is measured [Debevec and Malik, 1997, Grossberg and Nayar,
30 2004, Bioucas-Dias et al., 2012, Heylen et al., 2014]. After such an entry-level response, a simple
31 latent Tucker tensor may appear higher-rank in the observed domain.

32 A direct remedy is to increase Tucker rank, but this expands the core and factors throughout the
33 model. More flexible nonlinear tensor decoders based on kernels, Gaussian processes, or neural
34 networks can also help, but they often make the relation to the Tucker backbone less transparent [Xu
35 et al., 2011, Zhe et al., 2016, Liu et al., 2017, Saragadam et al., 2024]. We ask a narrower question:
36 can a small observation link, learned jointly with the Tucker factors, make a tight multilinear rank
37 budget more effective when the dominant residual is shared and entrywise?

We instantiate this idea with a polynomial link, a classical low-dimensional tool for modeling nonlinear responses [Volterra, 1959, Boyd and Chua, 1985]. Let

$$\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket$$

be a Tucker latent tensor. Instead of fitting the observation directly by $\mathcal{S}$, we model

$$\hat{\mathcal{Y}} = f_{\beta}(\mathcal{S}) = \sum_{p=0}^P \beta_p \mathcal{S}^{\odot p},$$

where powers are entrywise. We call the model Nonlinear Tucker Decomposition with Polynomial Links (NTD-PL). It contains Tucker as the degree-one case and adds only $P + 1$ scalar coefficients, while the powers $\mathcal{S}^{\odot p}$ induce tied high-order interactions among the same core and factors. Thus the link is not an independent high-rank correction; it is a compact way to expose nonlinear structure already organized by the Tucker backbone.

This viewpoint is central to the paper. NTD-PL is best understood as a Tucker enhancement and shared-backbone rank lift, not as a generic nonlinear decoder. A degree- p power of the latent tensor behaves like an ordinary Tucker tensor whose factors are polynomial feature lifts of the original factors, but those lifted features remain tied to one learned backbone. The new degrees therefore reshape the low-rank geometry through a shared observation link, rather than adding an untied high-rank tensor block.

Our claims are intentionally scoped. NTD-PL is not meant to replace rank selection or general nonlinear tensor decoders. It targets settings where a shared entry-level response explains coherent residual structure left by a low-rank Tucker model.

Contributions. We make the following contributions.

- We propose NTD-PL, a Tucker model with a jointly learned polynomial observation link. Standard Tucker is recovered as the degree-one case, while higher degrees act as a Tucker enhancement and shared-backbone rank lift of the same backbone.
- We characterize the model class through Veronese rank-expansion, separation from standard Tucker at the same backbone rank, and approximation results for shared nonlinear observation maps.
- We derive a masked learning objective for reconstruction and completion, and use an alternating solver with a closed-form ridge update for the link coefficients.
- We evaluate the mechanism on controlled tensors and hyperspectral data, including completion, reconstruction, rank-lift, and frozen-backbone residual-diagnostic checks, including predicted boundary cases.

2 Related Work

Low-rank tensor models. CP, Tucker, tensor train, and tensor ring decompositions provide compact representations for multiway data [Kolda and Bader, 2009, De Lathauwer et al., 2000, Oseledets, 2011, Zhao et al., 2016]. Tucker is the closest baseline for NTD-PL because its core and mode factors give the same multilinear backbone. Rather than changing ranks, factor structures, or regularization, we keep this backbone fixed and change the observation map from the latent tensor to the measured entries.

Nonlinear tensor factorization. Nonlinear tensor methods use generalized likelihoods, kernels, Gaussian processes, neural decoders, or nonlinear tensor networks [Hong et al., 2020, Xu et al., 2011, Zhe et al., 2016, Pan et al., 2020, Saragadam et al., 2024, Varolgunes et al., 2023]. These models can be more expressive than a scalar polynomial link, but they often trade away the simple finite-dimensional relation to Tucker. NTD-PL is complementary: it keeps the nonlinearity shared and entrywise so that its effect can be analyzed as a tied rank lift of the same Tucker factors.

Polynomial feature lifts and high-order interactions. Polynomial and Volterra-style expansions are classical tools for representing nonlinear responses [Volterra, 1959, Boyd and Chua, 1985]. Related matrix and tensor methods use polynomial feature maps or symmetric high-order tensors to

increase expressivity [Zhai et al., 2024, Liu et al., 2015, Deng et al., 2016]. Such lifts can expose nonlinear structure, but explicit high-order parameters often increase dimension and contraction cost. NTD-PL can be viewed as a shared-backbone compression of these interactions: the lifted features are induced by powers of one learned Tucker latent tensor, rather than by an independent high-order tensor parameter.

3 Model

3.1 Model and Observation Assumption

Let $\mathcal{Y} \in \mathbb{R}^{I_1 \times \dots \times I_N}$ be an observed tensor and let

$$\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket, \quad \mathcal{X} \in \mathbb{R}^{R_1 \times \dots \times R_N}, \quad \mathbf{A}^{(n)} \in \mathbb{R}^{I_n \times R_n}. \quad (1)$$

NTD-PL treats $\mathcal{S}$ as a shared low-rank latent tensor and models each observed entry through a scalar polynomial response:

$$Y_{\mathbf{i}} = f_{\beta}(S_{\mathbf{i}}) + \varepsilon_{\mathbf{i}}, \quad f_{\beta}(s) = \sum_{p=0}^P \beta_p s^p, \quad \mathbf{i} = (i_1, \dots, i_N). \quad (2)$$

Equivalently, the tensor prediction is

$$\hat{\mathcal{Y}} = f_{\beta}(\mathcal{S}) = \sum_{p=0}^P \beta_p \mathcal{S}^{\odot p}, \quad \beta \in \mathbb{R}^{P+1}. \quad (3)$$

The constant term β_0 captures affine response shifts, the linear term $\beta_1 \mathcal{S}$ is the usual Tucker channel, and terms with $p \geq 2$ activate nonlinear channels on the same latent tensor. Standard Tucker is recovered by setting $\beta_1 = 1$ and $\beta_p = 0$ for $p \neq 1$. Thus NTD-PL changes the entry-generation map while leaving the Tucker backbone and its multilinear ranks fixed.

3.2 Shared-Backbone High-Order Interactions

Although f_{β} is scalar, the term $\mathcal{S}^{\odot p}$ is not merely an independent output calibration. We arrive at it by first defining an explicit p -fold Tucker extension: instead of mapping one latent index per mode, the mode- n projection is a higher-order tensor $\mathcal{A}^{(n,p)} \in \mathbb{R}^{I_n \times R_n^p}$ that couples p latent indices. The case $p = 1$ is ordinary Tucker, while $p = 2$ gives a quadratic Tucker-style interaction between two latent Tucker tuples. Appendix A.1 gives the formal entrywise construction and tensor-network view.

The explicit construction is a reference model, not the parameterization we train. Its projection tensors grow as $I_n R_n^p$, and each degree would add a new set of mode projections. NTD-PL keeps the same high-order interaction pattern by tying the p -fold projection to the original Tucker factor:

$$\mathcal{A}_{i_n, r_n^{(1)}, \dots, r_n^{(p)}}^{(n,p)} = \prod_{q=1}^p A_{i_n, r_n^{(q)}}^{(n)}.$$

Under this tying, the degree- p channel is exactly $S_{\mathbf{i}}^p$. Thus it still couples p latent Tucker index tuples, but all copies share the same mode factors. Across degrees, the only new free parameters are the scalar coefficients $\{\beta_p\}$, which mix these tied channels.

The latent tensor $\mathcal{S}$ and the coefficients β are fitted jointly, so the low-rank backbone is optimized together with the polynomial channels. Section 4 analyzes the induced rank expansion and the resulting separation from standard Tucker.

3.3 Masked Least-Squares Objective

Let $\Omega \in \{0, 1\}^{I_1 \times \dots \times I_N}$ denote the observed-entry mask and let $|\Omega| = \sum_{\mathbf{i}} \Omega_{\mathbf{i}}$. We estimate the core, factors, and link coefficients by the regularized masked objective

$$\min_{\mathcal{X}, \{\mathbf{A}^{(n)}\}, \beta} \frac{1}{2|\Omega|} \|\Omega \odot (\mathcal{Y} - f_{\beta}(\mathcal{S}))\|_F^2 + \frac{\lambda_X}{2} \|\mathcal{X}\|_F^2 + \sum_{n=1}^N \frac{\lambda_n}{2} \|\mathbf{A}^{(n)}\|_F^2 + \frac{\lambda_{\beta}}{2} \|\beta\|_2^2. \quad (4)$$

The same formulation covers full reconstruction by taking $\Omega \equiv \mathbf{1}$ and tensor completion by restricting the loss to the observed entries. The quadratic penalties regularize the Tucker backbone and the low-dimensional polynomial link.

4 Theory

The theory addresses the rank-lift view raised by the model definition. A scalar link has few free parameters, but its powers can induce ordinary Tucker tensors with larger multilinear ranks. We first make this induced structure explicit, then give a separation from matched-rank Tucker and an approximation route for shared nonlinear observation maps.

4.1 Rank Expansion and Separation

Proposition 1 (Shared-backbone rank expansion). *Let $\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket$ have Tucker rank at most $(R_1, \dots, R_N)$. For every integer $p \geq 1$, $\mathcal{S}^{\odot p}$ admits a Tucker representation with mode ranks at most*

$$\left(\binom{R_1 + p - 1}{p}, \dots, \binom{R_N + p - 1}{p} \right). \quad (5)$$

More explicitly, the ordinary Tucker factors of $\mathcal{S}^{\odot p}$ can be chosen as Veronese lifts of the original factors. For $\mathbf{A} \in \mathbb{R}^{I \times R}$, define $\mathbf{A}^{(p)} \in \mathbb{R}^{I \times D}$, $D = \binom{R+p-1}{p}$, by

$$A_{i,\alpha}^{(p)} = \prod_{r=1}^R A_{i,r}^{\alpha_r}, \quad (6)$$

where $\alpha \in \mathbb{N}^R$ ranges over $|\alpha| = p$. Then $\mathcal{S}^{\odot p}$ factors through $\mathbf{A}^{(1)(p)}, \dots, \mathbf{A}^{(N)(p)}$.

This proposition makes the tying in Section 3.2 explicit in standard Tucker language: a degree- p channel is an ordinary Tucker tensor whose mode factors are polynomial feature lifts of the same backbone factors. The link therefore changes rank structure through tied features, not through an independent high-order parameter block.

Theorem 1 (Tensorial separation from standard Tucker). *Let $N \geq 3$, $R \geq 2$, $p \geq 2$, and suppose $I_n \geq D$ for every n , where $D = \binom{R+p-1}{p}$. Then there exists an order- N tensor $\mathcal{Y} \in \mathbb{R}^{I_1 \times \dots \times I_N}$ that is representable by a degree- p NTD-PL model with Tucker backbone rank $(R, \dots, R)$, but whose exact standard Tucker rank is*

$$(D, \dots, D). \quad (7)$$

Equivalently, a degree- p polynomial link can generate tensors whose multilinear ranks are $(\binom{R+p-1}{p}, \dots, \binom{R+p-1}{p})$ from a shared Tucker backbone of rank $(R, \dots, R)$.

The proof in Appendix B uses a CP-diagonal Tucker backbone and generic full-rank Veronese feature matrices. The theorem is an existence separation, not a universality claim over all high-rank Tucker tensors: it identifies a structured high-rank subset generated by a low-rank shared backbone.

4.2 Approximation Under Nonlinear Observations

Theorem 2 (Polynomial generation). *Suppose $\mathcal{Y} = g(\mathcal{S}^*)$, where $\mathcal{S}^*$ has Tucker rank at most $(R_1, \dots, R_N)$ and g is a polynomial of degree at most P . Then degree- P NTD-PL with the same Tucker rank can represent $\mathcal{Y}$ exactly.*

Theorem 3 (Analytic link approximation). *Suppose $\mathcal{Y} = g(\mathcal{S}^*)$, where $\mathcal{S}^*$ has Tucker rank at most $(R_1, \dots, R_N)$ and $\|\mathcal{S}^*\|_\infty \leq B$. If g is analytic on the complex disk $D(0, \rho)$ with $B < \rho$, and $M_\rho = \sup_{|z|=\rho} |g(z)|$, then there exists a degree- P NTD-PL model with the same Tucker backbone satisfying*

$$\frac{1}{\sqrt{M}} \|\mathcal{Y} - \hat{\mathcal{Y}}\|_F \leq M_\rho \frac{(B/\rho)^{P+1}}{1 - B/\rho}, \quad M = \prod_{n=1}^N I_n. \quad (8)$$

Thus the same finite link that induces a tied rank lift also gives a direct approximation route for smooth entrywise responses on a bounded latent range. The statement uses the simple Taylor truncation bound; sharper polynomial approximation bounds can replace it under stronger assumptions.

Together, the results clarify the scope of NTD-PL. Polynomial links can approximate shared nonlinear observation maps while keeping the same latent Tucker rank, and their powers induce Veronese-lifted Tucker factors. This explains how a scalar link can change multilinear rank structure without introducing untied high-order tensor parameters.

5 Optimization

We optimize the masked objective in Section 3.3 with a simple alternating scheme. The Tucker backbone $(\mathcal{X}, \{\mathbf{A}^{(n)}\})$ is updated by first-order steps through the current polynomial link, while the link coefficients are updated by a small ridge-regression problem on the current latent entries. This gives a practical solver while keeping the optimization layer standard.

Given the current latent tensor $\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket$, the coefficient update for active degree Q solves

$$\min_{\beta_{0:Q}} \frac{1}{2|\Omega|} \|\mathbf{y}_\Omega - \Phi_{\Omega,Q}(\mathcal{S})\beta_{0:Q}\|_2^2 + \frac{\lambda_\beta}{2} \|\beta_{0:Q}\|_2^2, \quad (9)$$

where $\Phi_{\Omega,Q}(\mathcal{S})$ contains the observed powers $[1, s, \dots, s^Q]$. The backbone update uses the chain rule through f_β ; the implementation evaluates $f_\beta(\mathcal{S})$ and $f'_\beta(\mathcal{S})$ together, then backpropagates the masked residual through the Tucker reconstruction.

Training starts from the Tucker-compatible degree-one model and activates higher powers gradually. When a new degree is activated, its coefficient is initialized at zero, so the model expands from the current low-degree fit rather than beginning with all nonlinear terms active. We also normalize factor scales and absorb them into the core, which fixes a Tucker gauge without changing $\mathcal{S}$.

The dominant cost is the same forward and reverse Tucker contractions used by standard Tucker optimization at the chosen rank. The polynomial link adds $O(P|\Omega|)$ elementwise work and a $(P+1) \times (P+1)$ ridge solve, so for the small degrees used here its overhead is minor. Full update equations, stabilization details, and pseudocode are given in Appendix A.2.

6 Experiments

The empirical study tests a scoped mechanism: when a Tucker backbone is kept low-rank, can a shared polynomial link act as a parameter-tied rank lift for residual structure that is invisible to a linear Tucker observation model? The experiments are organized to separate this question from a broad benchmark claim. Controlled tensors isolate when the link should help, masked CAFE completion tests whether the learned low-rank geometry generalizes to held-out entries, CAFE reconstruction checks parameter efficiency and joint linked fitting, and external HSI scenes probe transfer and predicted boundary cases.

Protocol. Unless stated otherwise, real tensors are globally max-normalized and used without spatial downsampling or cropping. NTD-PL is compared with Tucker at the same multilinear rank, except in the explicit rank-lift ablation where Tucker is given extra rank. We report RMSE and spectral angle (SAM). Completion masks entries uniformly at random and computes all completion metrics only on the held-out entries.

We also use a frozen-backbone diagnostic, the *residual-link score* S_{link} : after fitting Tucker, we refresh only the shared scalar link and score the corresponding residual reduction on the task-specific evaluation entries, $S_{\text{link}} = 10 \log_{10} \left(\frac{\|\mathcal{Y}_{\text{eval}} - \hat{\mathcal{Y}}_{\text{T}}\|_F^2}{\|\mathcal{Y}_{\text{eval}} - \hat{\mathcal{Y}}_{\text{link}}\|_F^2} \right)$. This keeps the diagnostic dimensionless and comparable across scene sizes and value scales, while giving a more legible score range than a near-zero energy fraction.

6.1 Controlled Nonlinear Tensors

The first check isolates the intended effect. We generate tensors of the form $Y_\alpha = \sqrt{1-\alpha} S^* + \sqrt{\alpha} \tilde{R}^*$, where S^* is low-rank Tucker and $\tilde{R}^*$ is the normalized component of a candidate nonlinear response that is orthogonal to S^* . Thus α controls nonlinear residual energy rather than the amplitude of a raw scalar nonlinearity.

When the source is linear, NTD-PL reduces to Tucker up to the fitted affine link; when nonlinear residual energy is present, the higher-order link terms should become useful. Figure 1 shows this pattern across three residual families: NTD-PL separates from linear tensor baselines as α increases. Figure 2 then varies the maximum polynomial degree at fixed residual energy, showing that the gain is tied to the expressive order of the link. Degree-continuation curves in Appendix C.1 show that

activating higher powers does not destabilize the fit. The formal residual construction and fitted link-term contributions are reported in Appendix C.1.

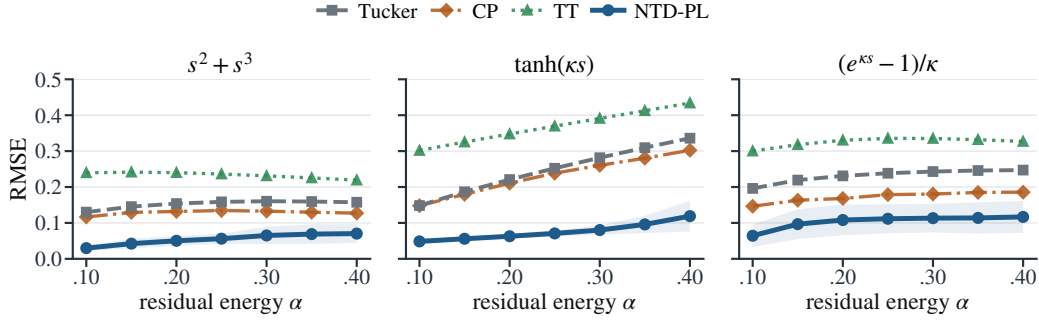


Figure 1: Controlled nonlinear alpha sweep. Each panel uses a different scalar response to define the orthogonal residual; lower RMSE is better.

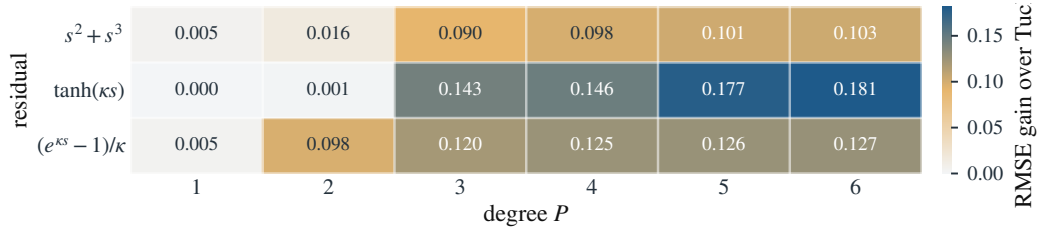


Figure 2: Controlled nonlinear degree-gain heatmap at $\alpha = 0.25$. Cell values are Tucker minus NTD-PL RMSE, so larger values favor NTD-PL.

6.2 Low-Rank Masked Completion

Masked completion is the main empirical test because held-out entries expose whether the learned low-rank geometry transfers beyond the observed tensor. In this setting, the polynomial link is useful only if it changes the inductive bias of the shared Tucker backbone, rather than simply calibrating training entries. Table 1 shows that this effect persists across random-mask rates at full resolution: with the same rank budget as Tucker, NTD-PL gives consistent held-out RMSE gains and also improves spectral angle on most scenes. The scene-level sign and Wilcoxon tests in Appendix C.2 support the same conclusion.

Table 1: Full-resolution CAFE completion at rank $(33, 33, 4)$ under random masks. Metrics are computed on missing entries. Tucker and NTD-PL columns report mean $\pm$ standard deviation over scenes; gains are paired relative improvements over Tucker.

ρ	RMSE* $\downarrow$			SAM* $\downarrow$		
	Tucker	NTD-PL	Gain	Tucker	NTD-PL	Gain
0.1	0.0263 $\pm$ 0.0179	0.0229$\pm$0.0158	12.9%	11.82 $\pm$ 5.33	8.36$\pm$3.38	29.3%
0.3	0.0263 $\pm$ 0.0179	0.0230$\pm$0.0158	12.5%	14.75 $\pm$ 6.33	12.47$\pm$5.06	15.5%
0.5	0.0264 $\pm$ 0.0179	0.0230$\pm$0.0158	12.9%	15.56 $\pm$ 6.85	13.48$\pm$5.75	13.4%
0.7	0.0266 $\pm$ 0.0180	0.0219$\pm$0.0128	17.7%	16.02 $\pm$ 7.21	13.79$\pm$5.43	13.9%

A cheap diagnostic supports the same interpretation. For each scene, we freeze the Tucker backbone and refresh only the shared scalar link, producing the residual-link score S_{link} . This is exactly one restricted NTD-PL substep with the latent Tucker geometry held fixed, so it isolates whether the matched-rank Tucker residual already contains a coherent shared scalar component. Table 2 shows the same pattern directly: across missing rates, higher-yield scenes receive larger NTD-PL gains on average, while the scene-level Spearman correlation remains strong throughout. Thus the boundary condition is empirically visible before full joint fitting: the link helps most when a frozen Tucker backbone already exposes a consistent scalar residual pattern.

Table 2: Residual-link diagnostic on CAVE completion. Low/mid/high gains report the mean NTD-PL RMSE gain over Tucker within each S_{link} tercile.

ρ	Spearman ρ_s	Low-yield gain	Mid-yield gain	High-yield gain	Scenes
0.1	0.829	0.75%	7.92%	12.63%	15
0.3	0.871	-1.13%	9.10%	13.12%	15
0.5	0.889	-1.09%	9.16%	13.18%	15
0.7	0.868	-0.62%	9.22%	13.07%	15

6.3 CAVE Reconstruction Mechanism Checks

Full-observation reconstruction is a controlled mechanism check rather than the main generalization test: with all entries observed, we can ask whether the polynomial link consistently removes residual structure left by a matched-rank Tucker backbone. Figure 3 supports this interpretation. NTD-PL improves both intensity error and spectral angle across the rank range, while the advantage becomes smaller as the Tucker rank increases. This trend is the expected signature of a lightweight observation model: it helps most when the linear backbone is tight, and becomes less necessary as ordinary multilinear capacity grows.

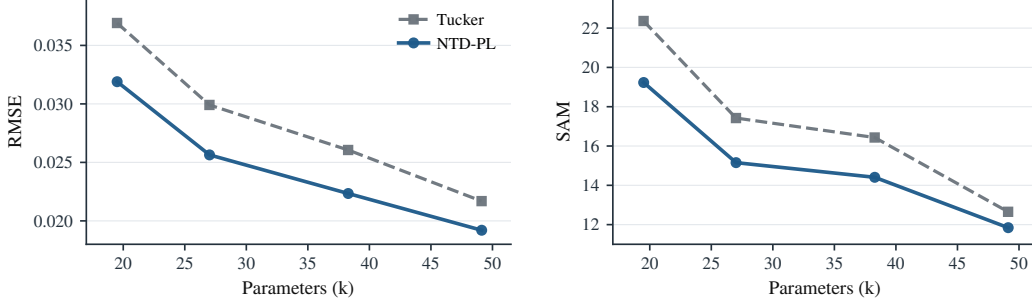


Figure 3: CAVE full-observation reconstruction as a representation-capacity check. Each point averages 15 scenes at a matched Tucker backbone rank. Unlike masked completion in Table 1, this plot measures how much error each parameter budget can represent when all entries are observed.

The next check clarifies why the reconstruction gain is a Tucker enhancement rather than a simple fixed-coefficient effect. First, Table 3 asks how much ordinary Tucker rank is needed before Tucker first matches the RMSE of a lower-rank NTD-PL model, while holding the spectral rank fixed. Tucker can buy back part of the RMSE gap by increasing rank, so NTD-PL should not be read as a replacement for rank selection. The spectral-angle gap, however, remains visible at the first RMSE-matching Tucker ranks. This supports the tied rank-lift view: NTD-PL reaches the same intensity error with many fewer parameters and preserves a different spectral geometry.

Table 3: Rank-lifted Tucker comparison on CAVE reconstruction. The table reports the first fixed-channel Tucker rank that matches each NTD-PL RMSE.

NTD-PL reference				First fixed-channel Tucker matching RMSE				Extra
Rank	Params	RMSE↓	SAM↓	Rank	Params	RMSE↓	SAM↓	
(18, 18, 3)	20k	0.0319	19.23	(28, 28, 3)	31k	0.0316	21.41	+59.5%
(24, 24, 4)	27k	0.0256	15.15	(35, 35, 4)	41k	0.0254	16.26	+51.3%
(33, 33, 4)	38k	0.0223	14.41	(49, 49, 4)	60k	0.0222	15.28	+56.5%

The same comparison also makes the computational tradeoff clear. The current first-order NTD-PL solver is slower than the highly optimized Tucker sweep, but this runtime cost is separate from the representational point: matching the NTD-PL RMSE with ordinary Tucker requires a substantially larger parameter budget and still leaves higher SAM.

6.4 External HSI Validity and Boundary Cases

Figure 4 turns the same completion setting into a scene-level diagnostic check at $\rho = 0.5$. The goal here is not another performance summary, but to test whether the proposed residual-link score S_{link} actually captures which scenes are more compatible with the shared-link mechanism. If the diagnostic is meaningful, scenes with larger scores should also exhibit larger NTD-PL gains, while scenes with weak scores should benefit less.

Table 4 asks whether the same mechanism transfers to other hyperspectral tensors and where the shared-link assumption weakens. Each dataset uses one matched low-capacity point chosen to keep both RMSE and gain in a reasonable regime. NTD-PL improves reconstruction on Jasper Ridge, Samson, and Urban, while Cuprite sits near the predicted boundary case: Tucker is already strong there, leaving little coherent residual for a shared polynomial link. This keeps the claim scoped: NTD-PL is most useful under a tight rank budget when the linear Tucker residual is coherent enough to be modeled by a shared nonlinear link. Additional cross-domain checks are in Appendix C.6.

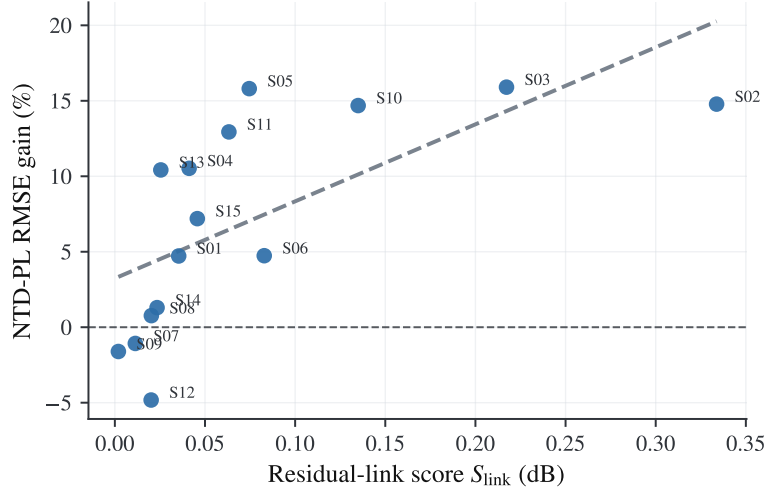


Figure 4: CAVE completion at $\rho = 0.5$: scene-level NTD-PL RMSE gain versus residual-link score S_{link} . Each point is one scene.

Table 4: External HSI reconstruction under matched low-capacity Tucker backbones. Columns report RMSE, relative RMSE gain over Tucker, and residual-link score S_{link} .

Dataset	CR	Tucker	NTD-PL	Gain	S_{link}
Jasper Ridge	124.4	0.0462	0.0430	6.93%	0.13
Samson	129.6	0.0263	0.0242	7.95%	0.06
Urban	126.3	0.0444	0.0428	3.59%	0.06
Cuprite	122.4	0.0176	0.0177	-0.60%	0.00

7 Limitations

NTD-PL assumes that the residual left by a low-rank Tucker backbone is well described by a shared entry-level nonlinear response. It is therefore not a universal tensor model: when Tucker already fits well at the chosen rank, or when residual nonlinearities vary strongly across space, spectra, or modes, a single scalar link may add little.

The method also does not remove rank selection. Increasing Tucker rank can recover part of the RMSE gap, and NTD-PL inherits Tucker’s nonconvexity and initialization sensitivity. Although the link adds few parameters, evaluating it over all observed entries still increases runtime relative to plain Tucker.

8 Conclusion

We introduced NTD-PL, a Tucker enhancement with a shared polynomial observation link. Rather than replacing multilinear low-rank structure, NTD-PL acts as a shared-backbone rank lift: powers of one learned latent tensor express structured residuals left by a linear Tucker fit without adding an independent high-rank correction.

The experiments support the same scoped interpretation. Controlled tensors, held-out CAVE completion, reconstruction mechanism checks, residual-link diagnostics, and external HSI scenes show that the link helps when a shared nonlinear residual is plausible under a tight rank budget, while also revealing predicted near-linear boundary cases. Adaptive links and broader tensor modalities are natural next steps.

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333 A Additional Model Details

334 A.1 Explicit p -Fold Tucker Motivation

335 This section expands the motivation behind the explicit p -fold Tucker channel in Section 3.2. The
336 construction is best viewed as the tensor analogue of adding polynomial interactions to a linear map.
337 In the matrix case, one may move from a linear model $\mathbf{y} = \mathbf{A}\mathbf{x}$ to a quadratic model

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathcal{A}(\mathbf{x}, \mathbf{x}),$$

338 where the second term couples pairs of latent coordinates. The corresponding question for Tucker is:
339 if standard Tucker is the multilinear low-rank map for multiway data, what is the direct p -th order
340 interaction analogue?

341 For a Tucker backbone, an explicit p -fold interaction can be written as

$$\tilde{Y}_{\mathbf{i}}^{(p)} = \sum_{\mathbf{r}^{(1)}, \dots, \mathbf{r}^{(p)}} \left(\prod_{q=1}^p \mathcal{X}_{r_1^{(q)}, \dots, r_N^{(q)}} \right) \prod_{n=1}^N \mathcal{A}_{i_n, r_n^{(1)}, \dots, r_n^{(p)}}^{(n,p)}. \quad (10)$$

342 Here each mode projection depends on p latent indices instead of one. For $p = 1$, this reduces to the
343 ordinary Tucker entry formula. For $p = 2$, the same core appears twice and the mode projections
344 couple two latent tuples, giving a quadratic Tucker-style channel. Figure 5 illustrates this progression.

345 The explicit construction is useful as a reference model, but it is not the parameterization used by
346 NTD-PL. Its mode projections $\mathcal{A}^{(n,p)}$ have size $I_n R_n^p$, so a direct implementation introduces new
347 high-order projection tensors and rapidly increases storage and contraction cost. NTD-PL keeps the
348 same high-order interaction pattern by tying the projection tensor to the original Tucker factor:

$$\mathcal{A}_{i_n, r_n^{(1)}, \dots, r_n^{(p)}}^{(n,p)} = \prod_{q=1}^p \mathcal{A}_{i_n, r_n^{(q)}}^{(n)}. \quad (11)$$

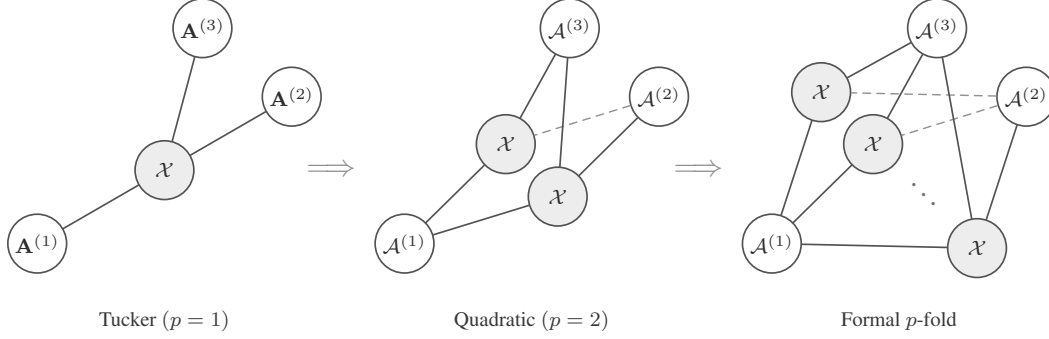


Figure 5: Tensor-network view of a formal p -fold Tucker extension. The construction keeps the Tucker idea of mode-wise generation, but lets each mode projection couple multiple latent index tuples.

349 Substituting (11) into (10) gives exactly $\mathcal{S}^{\odot p}$, where $\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket$. Thus the degree- p
 350 NTD-PL channel is a shared-projection version of the explicit p -fold Tucker prototype.

351 This is the modeling role of the p -fold Tucker prototype: it defines the high-order Tucker interaction
 352 that NTD-PL compresses. The final model does not train the untied tensors $\mathcal{A}^{(n,p)}$; it keeps a single
 353 Tucker backbone and adds only the polynomial coefficients $\{\beta_p\}$.

354 A.2 Optimization Details

355 This section gives the update equations behind the solver summarized in Section 5. The implementa-
 356 tion follows the same structure: one Tucker reconstruction per iteration, a fused evaluation of the
 357 polynomial and its derivative, first-order updates of the backbone, and periodic ridge-regression
 358 updates of the link coefficients.

359 **Backbone gradients.** For a fixed β , define

$$f'_\beta(\mathcal{S}) = \sum_{p=1}^P p\beta_p \mathcal{S}^{\odot(p-1)}. \quad (12)$$

360 The gradient of the normalized masked data term with respect to the latent tensor $\mathcal{S}$ is

$$\mathbf{T} = \frac{1}{|\Omega|} \Omega \odot (f_\beta(\mathcal{S}) - \mathcal{Y}) \odot f'_\beta(\mathcal{S}). \quad (13)$$

361 Backpropagating $\mathbf{T}$ through the Tucker reconstruction gives the core gradient

$$\nabla_{\mathcal{X}} \mathcal{L} = \mathbf{T} \times_1 \mathbf{A}^{(1)\top} \times_2 \dots \times_N \mathbf{A}^{(N)\top} + \lambda_X \mathcal{X}. \quad (14)$$

362 For the n -th factor, let

$$\mathbf{Z}^{(n)} = \left[\mathcal{X} \times_1 \mathbf{A}^{(1)} \dots \times_{n-1} \mathbf{A}^{(n-1)} \times_{n+1} \mathbf{A}^{(n+1)} \dots \times_N \mathbf{A}^{(N)} \right]_{(n)}. \quad (15)$$

363 Then

$$\nabla_{\mathbf{A}^{(n)}} \mathcal{L} = \mathbf{T}_{(n)} \mathbf{Z}^{(n)\top} + \lambda_n \mathbf{A}^{(n)}. \quad (16)$$

364 The experiments use Adam for these first-order steps. This choice is not part of the model definition;
 365 any comparable first-order optimizer can use the same gradients.

366 **Link update.** With $\mathcal{S}$ fixed and active degree $Q = P_{\text{act}}$, the coefficient subproblem is the ridge
 367 regression in (9). Its closed-form solution is

$$\beta_{0:Q}^* = (\Phi_{\Omega,Q}^\top \Phi_{\Omega,Q} + |\Omega| \lambda_\beta \mathbf{I})^{-1} \Phi_{\Omega,Q}^\top \mathbf{y}_\Omega. \quad (17)$$

368 Equivalently, one may absorb $|\Omega|$ into the ridge parameter and solve the usual unnormalized normal
 369 equations. If the constant term is disabled, the first column is removed and β_0 is fixed at zero. In
 370 the code, this small system is solved either from an explicit Vandermonde design or from moment
 371 equations built from cached powers.

372 **Stabilization and continuation.** High powers of $\mathcal{S}$ can be sensitive to scale. The implementation
 373 therefore normalizes factor columns after backbone updates and absorbs the corresponding scales
 374 into the core; this is a gauge fixing of the Tucker parameterization and does not change $\mathcal{S}$. For the
 375 coefficient solve, it may regress on the centered and scaled latent variable $(S - \mu)/\sigma$, then convert
 376 the solution back to the original power basis of S .

377 The active degree follows the schedule

$$t_k = \text{round} \left(\frac{kT}{P} \right), \quad k = 1, \dots, P-1.$$

378 When degree $p+1$ is activated, the previous coefficients and Tucker factors are kept and the new
 379 coefficient β_{p+1} is initialized at zero.

380 **Cost.** The dominant cost is forming the Tucker reconstruction and its reverse contractions on the
 381 observed tensor. For dense tensors this is comparable to standard Tucker optimization at the same
 382 rank, plus $O(P|\Omega|)$ work to evaluate the scalar polynomial and its derivative. The link update forms
 383 a Vandermonde design or the corresponding moment equations and solves a $(P+1) \times (P+1)$ linear
 384 system, with direct cost $O(P^2|\Omega| + P^3)$. Since P is small, this cost is negligible compared with the
 385 Tucker contractions.

386 Algorithm 1 gives the full procedural summary.

Algorithm 1 Masked NTD-PL optimization

Require: observed tensor $\mathcal{Y}$, mask Ω , rank $(R_1, \dots, R_N)$, degree P

- 1: initialize $\mathcal{X}, \{\mathbf{A}^{(n)}\}$ from Tucker on the masked tensor
 - 2: initialize active degree $P_{\text{act}} \leftarrow 1$ and $\beta = (0, 1)$
 - 3: **for** $t = 1, \dots, T$ **do**
 - 4: form $\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket$
 - 5: **if** a link update is scheduled **then**
 - 6: update active coefficients $\beta_{0:P_{\text{act}}}$ by ridge regression on $\{(S_i, Y_i) : \Omega_i = 1\}$
 - 7: **end if**
 - 8: compute $f_\beta(\mathcal{S})$ and $f'_\beta(\mathcal{S})$
 - 9: take first-order steps on $\mathcal{X}$ and $\{\mathbf{A}^{(n)}\}$ using the masked gradient
 - 10: normalize factor scales and absorb them into the core
 - 11: increase P_{act} according to the continuation schedule, initializing the new coefficient at zero
 - 12: **end for**
 - 13: **return** $\mathcal{X}, \{\mathbf{A}^{(n)}\}, \beta$
-

387 B Additional Proofs

388 B.1 Polynomial and Analytic Approximation

389 *Proof of Theorem 2.* Since g has degree at most P , write

$$g(s) = \sum_{p=0}^P c_p s^p. \quad (18)$$

390 Take the NTD-PL latent tensor to be the true latent tensor, $\mathcal{S} = \mathcal{S}^*$, and set $\beta_p = c_p$. Then, entrywise,

$$\hat{\mathcal{Y}}_{i_1, \dots, i_N} = \sum_{p=0}^P \beta_p (\mathcal{S}_{i_1, \dots, i_N}^*)^p = g(\mathcal{S}_{i_1, \dots, i_N}^*) = \mathcal{Y}_{i_1, \dots, i_N}. \quad (19)$$

391 Thus $\hat{\mathcal{Y}} = \mathcal{Y}$. □

392 **Lemma 1** (Scalar-to-tensor approximation). *Let $\|\mathcal{S}^*\|_\infty \leq B$. For any scalar function g and any*
 393 *degree- P polynomial q_P ,*

$$\|g(\mathcal{S}^*) - q_P(\mathcal{S}^*)\|_F \leq \sqrt{\prod_{n=1}^N I_n} \sup_{|s| \leq B} |g(s) - q_P(s)|. \quad (20)$$

394 *Proof.* Every entry of $\mathcal{S}^*$ lies in $[-B, B]$, so every entrywise approximation error is bounded by
 395 $\sup_{|s| \leq B} |g(s) - q_P(s)|$. Squaring, summing over all entries, and taking the square root gives the
 396 claim. $\square$

397 *Proof of Theorem 3.* Let $g(s) = \sum_{k=0}^{\infty} a_k s^k$ be the Taylor expansion of g around zero, and choose
 398 $q_P(s) = \sum_{k=0}^P a_k s^k$. By Cauchy's estimate,

$$|a_k| \leq \frac{M_\rho}{\rho^k}. \quad (21)$$

399 For $|s| \leq B < \rho$,

$$|g(s) - q_P(s)| \leq \sum_{k=P+1}^{\infty} |a_k| |s|^k \quad (22)$$

$$\leq M_\rho \sum_{k=P+1}^{\infty} (B/\rho)^k = M_\rho \frac{(B/\rho)^{P+1}}{1 - B/\rho}. \quad (23)$$

400 Applying Lemma 1 and taking $\mathcal{S} = \mathcal{S}^*$, $\beta_k = a_k$ for $0 \leq k \leq P$, yields the stated unnormalized
 401 bound. Dividing by $\sqrt{M}$, $M = \prod_{n=1}^N I_n$, gives the normalized form used in Theorem 3. $\square$

402 B.2 Rank Expansion

403 *Proof of Proposition 1.* Write the Tucker tensor entrywise as

$$\mathcal{S}_{i_1, \dots, i_N} = \sum_{r_1=1}^{R_1} \cdots \sum_{r_N=1}^{R_N} \mathcal{X}_{r_1, \dots, r_N} \prod_{n=1}^N A_{i_n, r_n}^{(n)}. \quad (24)$$

404 Taking an element-wise p -th power gives

$$(\mathcal{S}_{i_1, \dots, i_N})^p = \prod_{q=1}^p \left(\sum_{r_1^{(q)}=1}^{R_1} \cdots \sum_{r_N^{(q)}=1}^{R_N} \mathcal{X}_{r_1^{(q)}, \dots, r_N^{(q)}} \prod_{n=1}^N A_{i_n, r_n^{(q)}}^{(n)} \right) \quad (25)$$

$$= \sum_{\gamma_1 \in [R_1]^p} \cdots \sum_{\gamma_N \in [R_N]^p} \left(\prod_{q=1}^p \mathcal{X}_{r_1^{(q)}, \dots, r_N^{(q)}} \right) \prod_{n=1}^N \left(\prod_{q=1}^p A_{i_n, r_n^{(q)}}^{(n)} \right), \quad (26)$$

405 where $\gamma_n = (r_n^{(1)}, \dots, r_n^{(p)})$. For each mode n , let $\alpha_n \in \mathbb{N}^{R_n}$ record the counts of the entries in γ_n ,
 406 so $|\alpha_n| = p$. Then

$$\prod_{q=1}^p A_{i_n, r_n^{(q)}}^{(n)} = \prod_{r=1}^{R_n} \left(A_{i_n, r}^{(n)} \right)^{\alpha_{n,r}} = A_{i_n, \alpha_n}^{(n)\langle p \rangle}. \quad (27)$$

407 Group all ordered tuples $(\gamma_1, \dots, \gamma_N)$ that produce the same multi-indices $(\alpha_1, \dots, \alpha_N)$, and define
 408 the aggregated core

$$\mathcal{X}_{\alpha_1, \dots, \alpha_N}^{\langle p \rangle} = \sum_{\substack{\gamma_1, \dots, \gamma_N: \\ \text{count}(\gamma_n) = \alpha_n \ \forall n}} \prod_{q=1}^p \mathcal{X}_{r_1^{(q)}, \dots, r_N^{(q)}}. \quad (28)$$

409 The preceding display is exactly the Tucker reconstruction

$$\mathcal{S}^{\odot p} = \llbracket \mathcal{X}^{\langle p \rangle}; \mathbf{A}^{(1)\langle p \rangle}, \dots, \mathbf{A}^{(N)\langle p \rangle} \rrbracket. \quad (29)$$

410 The n -th Veronese factor has $\binom{R_n + p - 1}{p}$ columns, giving the claimed Tucker rank bound. $\square$

411 B.3 Separation

412 **Lemma 2** (Full-rank Veronese evaluations). *Let $D = \binom{R+p-1}{p}$. There exists a matrix $\mathbf{A} \in \mathbb{R}^{D \times R}$*
 413 *whose Veronese feature matrix $\mathbf{A}^{(p)} \in \mathbb{R}^{D \times D}$ has rank D . Consequently, for $I \geq D$, a generic*
 414 *matrix $\mathbf{A} \in \mathbb{R}^{I \times R}$ has $\text{rank}(\mathbf{A}^{(p)}) = D$.*

415 *Proof.* The homogeneous degree- p monomials in R variables are linearly independent polynomials.
 416 Hence there exist evaluation points $x_1, \dots, x_D \in \mathbb{R}^R$ such that the matrix $(x_i^\alpha)_{i,\alpha}$, $|\alpha| = p$, is
 417 nonsingular; otherwise every $D \times D$ evaluation determinant would vanish, which would imply a
 418 nontrivial linear dependence among the monomials. Taking these points as the rows of $\mathbf{A}$ gives the
 419 first claim.

420 For $I \geq D$, choose any D rows. The determinant of the corresponding $D \times D$ Veronese submatrix
 421 is a polynomial in the entries of $\mathbf{A}$ that is not identically zero by the first claim. Its nonvanishing is
 422 therefore a generic property, so $\mathbf{A}^{(p)}$ has full column rank for generic $\mathbf{A}$. $\square$

423 *Proof of Theorem 1.* Let $D = \binom{R+p-1}{p}$. Choose generic factor matrices $\mathbf{A}^{(n)} = [a_1^{(n)}, \dots, a_R^{(n)}] \in$
 424 $\mathbb{R}^{I_n \times R}$, and define the CP-diagonal Tucker backbone

$$\mathcal{S} = \sum_{r=1}^R a_r^{(1)} \otimes \dots \otimes a_r^{(N)}. \quad (30)$$

425 Its mode- n unfolding is

$$\mathcal{S}_{(n)} = \mathbf{A}^{(n)} \left(\mathbf{A}^{(N)} \odot \dots \odot \mathbf{A}^{(n+1)} \odot \mathbf{A}^{(n-1)} \odot \dots \odot \mathbf{A}^{(1)} \right)^\top. \quad (31)$$

426 For generic $\mathbf{A}^{(n)}$, $\mathbf{A}^{(n)}$ has rank R , and the Khatri-Rao product on the right has rank R . Thus
 427 $\text{rank}(\mathcal{S}_{(n)}) = R$ for every n , so $\mathcal{S}$ has exact Tucker rank $(R, \dots, R)$.

428 Set $\beta_p = 1$ and $\beta_q = 0$ for $q \neq p$. Then $\mathcal{Y} = \mathcal{S}^{\odot p}$ is a valid degree- p NTD-PL tensor. Entrywise,

$$\mathcal{Y}_{i_1, \dots, i_N} = \left(\sum_{r=1}^R \prod_{n=1}^N A_{i_n, r}^{(n)} \right)^p \quad (32)$$

$$= \sum_{|\alpha|=p} c_\alpha \prod_{n=1}^N \prod_{r=1}^R \left(A_{i_n, r}^{(n)} \right)^{\alpha_r}, \quad (33)$$

429 where $c_\alpha = p! / (\alpha_1! \dots \alpha_R!) > 0$. Let $\mathbf{B}^{(n)} = \mathbf{A}^{(n)\langle p \rangle} \in \mathbb{R}^{I_n \times D}$. By Lemma 2, each $\mathbf{B}^{(n)}$ has full
 430 column rank D for generic $\mathbf{A}^{(n)}$. The mode- n unfolding is

$$\mathcal{Y}_{(n)} = \mathbf{B}^{(n)} \text{diag}(c) \left(\mathbf{B}^{(N)} \odot \dots \odot \mathbf{B}^{(n+1)} \odot \mathbf{B}^{(n-1)} \odot \dots \odot \mathbf{B}^{(1)} \right)^\top. \quad (34)$$

431 The diagonal matrix $\text{diag}(c)$ is nonsingular. Since a Khatri-Rao product of full-column-rank matrices
 432 is full column rank, the right factor has rank D . Therefore $\text{rank}(\mathcal{Y}_{(n)}) = D$. This holds for every
 433 mode n , and the exact Tucker rank of $\mathcal{Y}$ is $(D, \dots, D)$. $\square$

434 B.4 Masked Generalization

435 **Theorem 4** (Masked-entry uniform convergence). *Let $\Theta \subset \mathbb{R}^d$ be a bounded NTD-PL parameter set*
 436 *with*

$$d = \prod_{n=1}^N R_n + \sum_{n=1}^N I_n R_n + (P+1).$$

437 For $\theta \in \Theta$, define the full-entry risk

$$\mathcal{L}(\theta) = \frac{1}{M} \sum_{\mathbf{i}} \ell_{\mathbf{i}}(\theta), \quad M = \prod_{n=1}^N I_n,$$

438 and the masked empirical risk

$$\widehat{\mathcal{L}}_m(\theta) = \frac{1}{m} \sum_{t=1}^m \ell_{\mathbf{i}_t}(\theta),$$

439 where $\mathbf{i}_t$ are sampled uniformly from tensor entries. Assume $0 \leq \ell_{\mathbf{i}}(\theta) \leq B_\ell$ and $\ell_{\mathbf{i}}(\theta)$ is G -Lipschitz
 440 in θ , uniformly over $\mathbf{i}$. If Θ is contained in a Euclidean ball of radius B_Θ , then with probability at
 441 least $1 - \delta$,

$$\sup_{\theta \in \Theta} \left| \mathcal{L}(\theta) - \widehat{\mathcal{L}}_m(\theta) \right| \leq 2G\varepsilon + B_\ell \sqrt{\frac{d \log(3B_\Theta/\varepsilon) + \log(2/\delta)}{2m}}$$

442 for every $0 < \varepsilon \leq B_\Theta$.

443 *Proof of Theorem 4.* Fix $0 < \varepsilon \leq B_\Theta$, and let $\mathcal{N}_\varepsilon$ be an ε -net of Θ in Euclidean norm. Since Θ lies
 444 in a d -dimensional ball of radius B_Θ , it admits a net with

$$|\mathcal{N}_\varepsilon| \leq \left(\frac{3B_\Theta}{\varepsilon} \right)^d. \quad (35)$$

445 For any fixed $\theta_0 \in \mathcal{N}_\varepsilon$, the random variables $\ell_{\mathbf{i}_t}(\theta_0)$ are independent and bounded in $[0, B_\ell]$.
 446 Hoeffding's inequality gives

$$\Pr \left(\left| \mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0) \right| \geq u \right) \leq 2 \exp \left(-\frac{2mu^2}{B_\ell^2} \right). \quad (36)$$

447 Taking a union bound over $\mathcal{N}_\varepsilon$, with probability at least $1 - \delta$,

$$\max_{\theta_0 \in \mathcal{N}_\varepsilon} \left| \mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0) \right| \leq B_\ell \sqrt{\frac{\log(2|\mathcal{N}_\varepsilon|/\delta)}{2m}}. \quad (37)$$

448 Substituting the net-size bound yields the stated stochastic term.

449 Now take any $\theta \in \Theta$, and choose $\theta_0 \in \mathcal{N}_\varepsilon$ with $\|\theta - \theta_0\|_2 \leq \varepsilon$. By the uniform Lipschitz assumption,

$$|\mathcal{L}(\theta) - \mathcal{L}(\theta_0)| \leq G\varepsilon, \quad |\widehat{\mathcal{L}}_m(\theta) - \widehat{\mathcal{L}}_m(\theta_0)| \leq G\varepsilon. \quad (38)$$

450 Therefore

$$\left| \mathcal{L}(\theta) - \widehat{\mathcal{L}}_m(\theta) \right| \leq |\mathcal{L}(\theta) - \mathcal{L}(\theta_0)| + \left| \mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0) \right| \quad (39)$$

$$+ |\widehat{\mathcal{L}}_m(\theta_0) - \widehat{\mathcal{L}}_m(\theta)| \quad (40)$$

$$\leq 2G\varepsilon + B_\ell \sqrt{\frac{d \log(3B_\Theta/\varepsilon) + \log(2/\delta)}{2m}}. \quad (41)$$

451 Taking the supremum over $\theta \in \Theta$ completes the proof. $\square$

452 **Remark 1.** The Lipschitz and bounded-loss assumptions are standard for compact parametric
 453 classes. For NTD-PL, they follow when the observed values, core, factor matrices, and polynomial
 454 coefficients are bounded and the polynomial degree P is finite. The resulting Lipschitz constant may
 455 depend polynomially on these bounds and on P , but the covering dimension remains the number
 456 of tied NTD-PL parameters rather than the number of parameters in an untied high-order Tucker
 457 expansion.

458 C Additional Experiments

459 C.1 Controlled Synthetic Mechanism Checks

460 The controlled experiments in the main text are designed to isolate the role of the scalar link. Table 5
 461 first checks a negative control: when the source is linear Tucker, NTD-PL does not improve by
 462 inventing nonlinear terms. The only exception is an affine offset, where the constant link term is the
 463 intended correction.

For the nonlinear controlled tensors, the nonlinearity is introduced as an orthogonal residual rather than by directly setting $Y = g(S^*)$. Let S^* be the linear low-rank Tucker backbone and let g be one of the candidate scalar responses. We form

$$R^* = g(S^*) - \frac{\langle g(S^*), S^* \rangle}{\|S^*\|_F^2} S^*, \quad (42)$$

and, when $\|R^*\|_F > 0$, normalize it as

$$\tilde{R}^* = \frac{\|S^*\|_F}{\|R^*\|_F} R^*. \quad (43)$$

The observed tensor is then

$$Y_\alpha = \sqrt{1 - \alpha} S^* + \sqrt{\alpha} \tilde{R}^*, \quad 0 \leq \alpha \leq 1. \quad (44)$$

By construction, $\langle S^*, \tilde{R}^* \rangle = 0$ and $\|\tilde{R}^*\|_F = \|S^*\|_F$, so α is exactly the energy fraction assigned to the nonlinear residual. This keeps the total energy fixed while moving the observation away from the linear Tucker backbone. The candidate residual families are $g(s) = s^2 + s^3$, $g(s) = (\exp(\kappa s) - 1)/\kappa$, and $g(s) = \tanh(\kappa s)$, with κ set from the input range.

The main text reports the nonlinear alpha sweep and degree sweep. Figure 6 shows that degree continuation activates higher powers without destabilizing the fit. Table 6 complements those figures with a small but consistent term-mass concentration effect: the dominant contribution usually lands on the degree that matches the injected response, while smooth odd links spread mass across nearby odd orders.

Table 5: Linear consistency on synthetic Tucker tensors. The $p > 1$ columns report effective higher-order NTD-PL contribution.

Method	bias = 0		bias = 0.5	
	RMSE↓	$p > 1$	RMSE↓	$p > 1$
Tucker	0.0114	—	0.0647	—
NTD-PL, $\beta_0 = 0$	0.0116	4.19%	0.0522	39.17%
NTD-PL, β_0 free	0.0117	3.19%	0.0118	2.98%

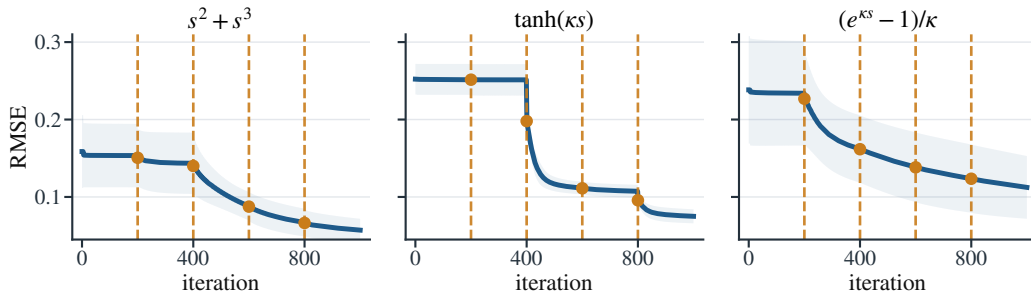


Figure 6: NTD-PL training curves for the controlled nonlinear mechanism checks at $\alpha = 0.25$ and $P = 5$. Dashed vertical lines mark degree-activation events.

Table 6: Representative effective NTD-PL term contributions under moderate nonlinearity, reported as percentages over ten seeds. Polynomial targets concentrate mass at the matching nonlinear degrees, while odd smooth links place most nonlinear mass on odd powers.

Link	Effective contribution (%)						
	$p = 0$	$p = 1$	$p = 2$	$p = 3$	$p = 4$	$p = 5$	$p = 6$
s^2	0.5	43.3	23.2	5.8	8.2	9.1	9.8
$s^2 + s^3$	0.1	24.1	9.8	38.1	7.3	11.5	9.1
$\sin(\kappa s)$	0.1	37.7	0.6	35.4	2.8	19.8	3.6
$\tanh(\kappa s)$	0.2	37.9	1.3	30.9	5.4	17.3	7.0

C.2 CAVE Completion Significance and Robustness

The main text reports the full-resolution random-missing CAVE completion table at rank (33, 33, 4). We do not repeat that table here; instead, Table 7 gives the paired sign and Wilcoxon tests for the same protocol.

Table 7: Significance tests for full-resolution CAVE completion at rank (33, 33, 4). Metrics are computed on missing entries.

ρ	RMSE				SAM			
	W/L	Med. gain	Sign p	Wilcoxon p	W/L	Med. gain	Sign p	Wilcoxon p
0.1	14/1	0.0027	9.77×10^{-4}	1.22×10^{-4}	14/1	2.9742	9.77×10^{-4}	8.54×10^{-4}
0.3	14/1	0.0027	9.77×10^{-4}	1.22×10^{-4}	13/2	1.9194	0.007	0.004
0.5	14/1	0.0026	9.77×10^{-4}	1.22×10^{-4}	13/2	1.6566	0.007	0.012
0.7	14/1	0.0027	9.77×10^{-4}	1.22×10^{-4}	12/3	1.3394	0.035	0.048

C.3 Residual-Link Diagnostic

The diagnostic used in the main text is a restricted NTD-PL update applied to a fixed Tucker backbone. Let $\hat{\mathcal{Y}}_T$ denote the Tucker output and $\mathcal{Y}$ the target tensor. We freeze the Tucker core and factors, and update only the scalar polynomial coefficients by fitting

$$h_\gamma(x) = \sum_{p=0}^4 \gamma_p x^p$$

by ridge regression on pairs $(\hat{Y}_{T,i}, Y_i)$. In full reconstruction, all entries are used for this fit. In completion, the fit uses only observed entries and is then evaluated on held-out entries. The fitted map is applied entrywise to $\hat{\mathcal{Y}}_T$; the Tucker core and factors are never updated. We summarize this refresh by the residual-link score

$$S_{\text{link}} = 10 \log_{10} \left(\frac{\|\mathcal{Y}_{\text{eval}} - \hat{\mathcal{Y}}_T\|_F^2}{\|\mathcal{Y}_{\text{eval}} - \hat{\mathcal{Y}}_{\text{link}}\|_F^2} \right),$$

which measures, on a log scale, how much Tucker residual energy is removed on the task-specific evaluation entries. In other words, it measures how much of the NTD-PL benefit is already visible before the backbone is allowed to move.

C.4 Joint Link-Family Smoke Test

The main model uses a polynomial link because it gives a closed-form ridge coefficient update and a direct Veronese rank-lift interpretation. To check whether the improvement is tied to the polynomial basis itself, we ran a small joint link-family smoke test on three CAVE scenes at spatial resolution 256×256 and backbone rank (18, 18, 3). The spline variant replaces the polynomial link by a one-dimensional piecewise-linear link with eight knots. Its knot values are updated by ridge regression on the current latent entries, while the Tucker backbone is updated jointly through the link derivative.

Table 8: Joint scalar-link family smoke test on CAVE reconstruction. The spline link is slightly more flexible but slower; NTD-PL is the analyzable polynomial member used in the main paper.

Method	Params	RMSE↓	SAM↓	Time↓
Tucker	10k	0.0542 ± 0.0401	24.47 ± 10.95	$1.4 \pm 0.1\text{s}$
NTD-PL(P=4)	10k	0.0502 ± 0.0363	23.23 ± 10.85	$10.7 \pm 0.1\text{s}$
Joint Spline Link(K=8)	10k	0.0497 ± 0.0362	20.25 ± 8.01	$40.4 \pm 0.4\text{s}$

C.5 CAVE Beads Pseudo-RGB Visualization

Figure 7 visualizes the CAVE beads scene under the reconstruction and random-completion protocols. Each 31-band tensor is rendered as pseudo-RGB by selecting three representative spectral bands near 75%, 50%, and 20% of the wavelength index range and applying a shared linear intensity scaling within each task. For completion, unobserved entries in the observation panel are shown in gray.

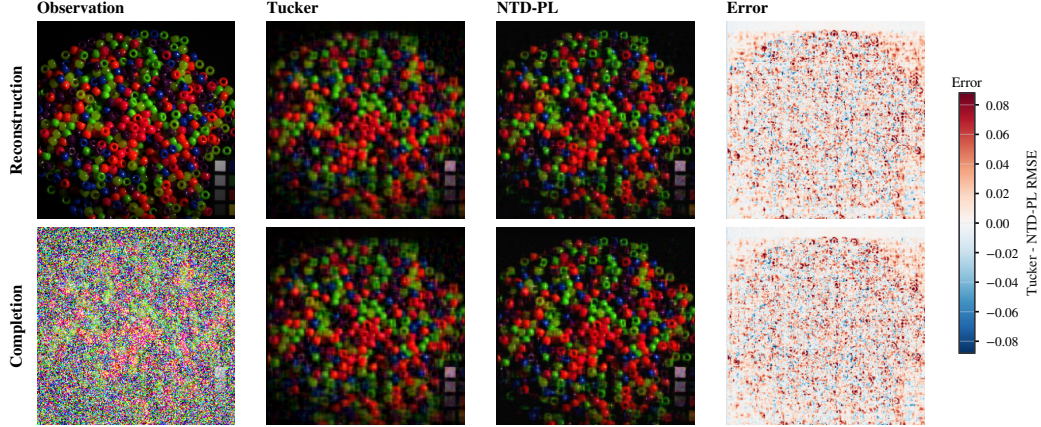


Figure 7: CAVE beads visual comparison. Rows correspond to reconstruction and random completion at missing rate 0.5. Error maps show Tucker minus NTD-PL per-pixel RMSE, so positive values indicate lower NTD-PL error.

C.6 Cross-Domain Real Tensor Checks

Table 9 reports matched-rank reconstruction on tensors outside the main hyperspectral setting: additional hyperspectral scenes, natural-image collections, and object-view tensors. The results are consistent with the scoped claim in Section 6.4: gains are clearest on hyperspectral and object-view data, while natural-image collections provide more modest improvements.

Table 9: Additional matched-rank real tensor reconstruction. Columns report RMSE; gains are relative RMSE reductions over Tucker, and positive Δ RMSE favors NTD-PL.

Domain	Dataset	Shape	Rank	RMSE↓			Gain
				Tucker	NTD-PL	Δ	
Hyperspectral	Jasper Ridge	$100 \times 100 \times 198$	(11,11,43)	0.0462	0.0430	0.0032	6.9%
Hyperspectral	Samson	$95 \times 95 \times 156$	(10,10,35)	0.0263	0.0242	0.0021	7.9%
Hyperspectral	Urban	$307 \times 307 \times 162$	(43,43,47)	0.0444	0.0428	0.0016	3.6%
Natural images	CBSD68	$68 \times 96 \times 96 \times 3$	(20,32,32,3)	0.1327	0.1307	0.0020	1.5%
Natural images	CIFAR-10	$1000 \times 32 \times 32 \times 3$	(96,20,20,3)	0.0753	0.0724	0.0029	3.8%
Object views	COIL-100	$8 \times 36 \times 48 \times 48 \times 3$	(4,8,12,12,2)	0.0991	0.0931	0.0060	6.0%
Object views	smallNORB	$25 \times 18 \times 64 \times 64 \times 2$	(18,12,24,24,2)	0.0454	0.0429	0.0024	5.4%

NeurIPS Paper Checklist

This draft uses a working checklist. It should be replaced by the official NeurIPS checklist distributed with the final 2026 submission template.

- Claims.** The main claims are stated in the abstract, introduction, theory section, and experiment captions: NTD-PL is a Tucker enhancement and shared-backbone rank lift for Tucker tensors; its gains are tied to jointly reshaping the Tucker backbone under a shared scalar link rather than to modest rank increases alone.
- Limitations.** Section 7 discusses predicted near-linear boundary cases, whole-axis missingness, shared-link assumptions, and the current domain concentration in hyperspectral data.

- 520 3. **Theory assumptions.** The approximation results assume a bounded latent tensor and polynomial
521 or analytic scalar links. The separation results use generic full-rank feature constructions. The
522 masked generalization bound assumes bounded parameters, bounded observations, and random
523 observed entries.
- 524 4. **Reproducibility.** The experiment section reports datasets, ranks, mask types, missing rates,
525 and baseline definitions. The codebase includes scripts for rank sweeps, MLPCal, structured
526 missingness, and the cross-domain real-tensor reconstruction study; the final version should add
527 hardware details and exact command manifests.